▣ 07 Bayes Theorem

Recall that Pr(A | B) denotes the probability of event A occurring, given than B has occurred.
It is worth considering a few examples to consolidate exactly what this means,

Example

Consider again the statistics exam taken by 200 students (opposite).
For a randomly selected student we can readily calculate the probability of the student being in any cell in the table. E.g.

	Engineering	Science	Business
Pass	48	49	63
Fail	12	21	7

Pr(Science student passing) = 49/200 = 0.245
Pr(Business student failing) = 7/200 = 0.035

By dividing each cell in the table by 200 we obtain probabilities.
Included are the marginal totals

	Engineering	Science	Business	
Pass	0.24	0.245	0.315	0.8
Fail	0.06	0.105	0.035	0.2
	0.3	0.35	0.35	

Define the events
P: Student passes the exam
S: Student is a science student

Clearly
Pr(P) = 0.240 + 0.245 + 0.315 = 0.8
Pr(S) = 0.245 +0 .105 = 0.35

Note that P ∩ S is the event that a student passes the exam AND studies science.
We can describe this composite event more concisely as - a science student passes.

What about Pr(P | S) - If a student is a science student, how likely is it that they passed?

That we are "given S" allows us to discount everything pertaining to engineering or business students – we can focus solely on the second, science column.

	Engineering	**Science**	Business	
Pass	*0.24*	**0.245**	0.315	0.8
Fail	*0.06*	**0.105**	0.035	0.2
	0.3	**0.35**	0.35	

If 0.35 is a measure of the total (for science) and 0.245 is a measure of just the science students that pass then initiatively, it would seem that the ratio $^{0.245}/_{0.350}$ = 0.7, or 70%, is the probability we required,

(We can perhaps more readily see that this is correct by carrying out a similar analysis in the original table – i.e., there are a total of 49 + 21 = 70 students and 49 of these or 49/70 = 0.7 (70%) pass)

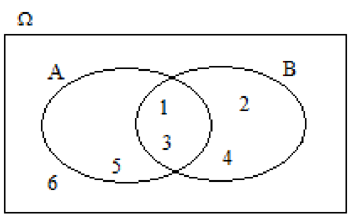
Note that 0.245 is the probability of a student being a science student AND passing = Pr(P ∩ S)
Also 0.35 is the probability of a student being a science student = Pr(S)

Thus, we can write Pr(P | S) = Pr(P ∩ S) / Pr(S)

This is <u>Bayes theorem</u> and holds for any two events.

Note that we can re-write this as Pr(P ∩ S) = Pr(P | S) Pr(S)

It can be useful to consider a trivial example to get a feel for why this result is correct. Consider again the die roll problem and the events A: outcome is odd and B outcome is 4 or less
Note that "given B" or | B means that B has definitely occurred.



So, the result must be either 1, 2, 3 or 4 and so for A also to occur requires that the outcome be either '1' or '3' – i.e., be in A ∩ B. Thus, we have Pr(A | B) = Pr(A ∩ B) / Pr(B)

Exercise
Find Pr(B|A) in the example above

Earlier we found Pr(P | S) = 0.7 = i.e., the probability of a student passing given that they study science.

What about Pr(S | P) - If a student passes, how likely is it that they were a science student?

That we are "given P" allows us to discount everything pertaining to students who fail – we can focus solely on the first, "pass" row.

	Engineering	Science	Business	
Pass	**0.24**	**0.245**	**0.315**	**0.8**
Fail	*0.06*	*0.105*	*0.035*	*0.2*
	0.3	0.35	0.35	

If 0.8 is a measure of the total (for pass) and 0.245 is a measure of just the science students that pass then initiatively, it would seem that the ratio $^{0.245}/_{0.8}$ = 0.306 is the probability we required,

But as before, .245 is the probability of a student being a science student AND passing = Pr(P ∩ S)
And 0.8 is the probability of a student passing = Pr(P)

So, we have Pr(S | P) = Pr(S ∩ P) / Pr(P)
Or Pr(S ∩ P) = Pr(S | P)Pr(P)

Again, Bayes theorem.

The complement law of probability and Bayes theorem

Note that Pr(¬A| B) + Pr(A | B) = 1. However. Pr(A | B) and Pr(A |¬B) usually have <u>no</u> relationship.

To illustrate, consider the example where 90% of a prison population and 50% of the general population are male.

Define the events P and M to be that a person is in prison and a person is male respectively.

Pr(M | P) + Pr(¬M| P) = 0.9 + 0.1 = 1
I.e., the percentage of males in prison + the percentage of females in prison must total 100%

Pr(M | P) + Pr(M |¬P) = 0.9 + 0.5 = 1.4 which has no meaning as a probability.

I.e., We are trying to add the percentage of males in prison + the percentage of males in the population!

<u>Problem 1</u>
On a housing estate of 100 houses, 40 are terraced, 50 are semi-detached and 10 are detached. 30 of the terrace houses, 40 of the semi-detached and 5 of the detached houses are rented – the rest are owner-occupied.
For a randomly selected house, define the events T (terraced), S (semi), D (detached) and R (rented)

We will consider a number of events, express them using the event definitions above and find their probabilities. In relatively simply problems, we can immediately see that our calculations are correct.

Exercise
(a) How likely is it that a semi-detached house is rented?
(b) How likely is it that a rented house is semi-detached?
(c) How likely is it that a rented house is **not** detached?

Exercise
In a sample of 50 laptop users, 35 use a PC (windows), the rest use a Mac and 10 use both. 28 of the PC users and 10 of the Mac users have anti-virus software installed as do 8 of the 10 who use both systems. One of the 50 is randomly selected.

If we define (for the selected user) the events W: uses a PC, V: has anti-virus software installed, express the following in terms of these events and find their probabilities.

(Note that ¬ W : is a mac user and ¬ V: does not have anti-virus software

(a) Is a PC user
(b) Is a Mac user
(c) Has anti-virus software
(d) Is a PC user and does **not** have anti-virus software
(e) Is a Mac user given that they have anti-virus software
(f) Had anti-virus software given that they are mac users
(g) Is a PC user given that they do **not** have anti-virus software

<u>Problem 2</u>
The solicitors in a large law firm are 40% female. 5% of females and 15% of males are partners.

(a) What percentage of the total work force are partners?
(b) What percentage of partners are female?

Define the events F: solicitor is female and P: solicitor is

Note that ¬ F: solicitor is male and ¬ P: solicitor is not a partner

5% of females are partners? Is this Pr(F | P) or Pr(P | F)

It is helpful to consider the preposition "of" when seeking to distinguish between the two above.
I.e., 5% ***of females*** *are partners: the "of females" indicates that we are considering the set made up solely of females, or we are restricting our interest to just females – thus it is Pr(P |F)*

Part (b) asks: What percentage of partners are female – here we are considering the set of made up solely of partners, or we are restricting our interest to just partners– thus it is Pr(F|P)

Note that we arbitrarily defined the event F to be a solicitor is female so ¬ F is a solicitor is male. We would get the same result if we defined M: a solicitor is male and thus ¬ M : a solicitor is female. (*Exercise)*

<u>Problem 3</u>
A spam filter is incorporated into an email client. The filter will correctly identify an email as spam 98% of the time but will erroneously mark as spam, a legitimate email 3% of the time. If a particular email user knows that 1% of all of the email, they receive are spam, what percentage of emails for this used marked as spam by the filter are in fact spam emails?

The problem above can also (and perhaps more easily) be solved by consider a sample of 10, 000 (or any number) of emails and using a tree diagram.

<u>Problem 4</u>
Two production lines in a factory X and Y produced 3% and 5% defectives respectively and account for all the factory output between them with X accounting for 60% of this. If an item is found to be defective, how likely is it that it was produced online X?